

System Monitoring

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Lectures

- 1 System administration introduction
- 2 Operating System installation
- 3 User management
- 4 Application management
- 5 **System monitoring**
- 6 Filesystem Maintenance
- 7 Network services
- 8 Security and Protection
- 9 Introduction to Public Cloud

Outline

- 1 Introduction
 - Goals
- 2 System Monitoring
- 3 Process management
- 4 User monitoring
- 5 Network monitoring

Goals

Knowledge

- Monitoring commands
- Meaning of the different signals

Abilities

- Obtain information about the system's behavior
 - CPU activity
 - Memory activity
 - Disk activity
- Process status monitoring
 - Priority change
 - Stop and Continue processes

Outline

- 1 Introduction
- 2 System Monitoring
 - CPU
 - GPU
 - Memory
 - Disk
 - Network
 - Users
 - Other monitoring tasks
- 3 Process management
- 4 User monitoring

System Monitoring

Why monitoring?

- Proactively control the resource status
- Control service status
- Security

Actions

- Automatic
- Manual

System Monitoring

What do we monitor?

- CPU
- GPU
- Memory
- I/O
- Network
- Users
- Services
- Logs

System Monitoring

Other factors

- When a resource is monitored?
- Who do we contact in case there is a problem?
- Which is the criteria to notify a warning?
- And for a critical issue?

CPU Activity

Monitoring

- Illegitimate applications
- Monopolized processors
 - By a single process
 - By a single user

Load Average

- Average number of processes competing for CPU: last 1, 5 and 15 minutes
- Allows to see the relative load of the system depending on the available number of cores in the system

Tools

`uptime, top, ps`

GPU Activity

Monitoring

- Applications hogging the GPU processing
- GPU Memory utilization
- Power usage

Tools

`nvidia-smi`, `glances`

Memory activity

Monitoring

- Lack of memory
- Memory monopolization
 - By a single process
 - By a single user
- Swap usage

Tools

`free, vmstat, top`

I/O Activity

Monitoring

- Filesystem
- Anomalous I/O activity
- Virtual memory
 - Excessive Pagination
 - Free Space

Importance

- Disks are generally the slowest path in our tasks
- Having an overloaded disk causes waittime → general slowness

Tools

iotop, df, du

Network Activity

Monitoring

- Bandwidth
- Local and remote services
- Incoming/outgoing connections
- Traffic profile
- Denial of Service (DoS)
- Distributed Denial of Service (DDoS)
- Service abuse
- Abroad incoming connections

Tools

`ip -s -d, ss, tcpdump, nmap, system logs`

User activity

Monitoring

- Active sessions
 - Locally
 - Remotely
- Connected users
- What are they doing?

Tools

w, last, fuser, lsof

Other monitoring tasks

Service and server activity

- Web server load
- E-mail queues
 - Input
 - Output
- Printer queues

Log files and journal

- System errors
- Service status
- Anomalous activity (security)
- Unexpected behaviour

Tools

journalctl, Wazuh, Crowd Strike Falcon

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 - Priority change
 - Signals
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Tasks and process management

Process identification

- Who is the owner of the process?
- Which is its purpose?
 - Is it important?
 - Is it an attack? ... or an error?

Actions on the process

- Priority changes
- Stop and reactivation of a process
- Killing a process

Priority change

- When executing the process
 - `nice +10 command...`
- Once it is already running
 - `renice +10 <pid>`
- Only root can increase the priority

Negative values indicate higher priorities

Some advise

High priority Shell

- Higher priority than swap
 - Allows a more efficient detection/solving of a memory issue
- The child processes inherit the priority of the parent

Relative priorities

- Priority is a relative term
- Not useful if all the processes have high priority

Sending signals to processes

```
kill <signal> <pid>
```

- -KILL: immediately stops the process
- -TERM: ask a process to gracefully finish (kill, by default)
- -INT: interrupt a process (kill, by default)
- -STOP: stop a process
 - Do not allow it to be enqueued in the ready queue
- -CONT: reactivate the selected process

```
killall <signal> <command name>
```

- Sends the signal to **ALL** the processes matching the name

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 - Examples
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User monitoring

User activity

- `w [user]`
 - List of connected users and the command being executed
 - Given a username, it lists his/her connections
- `last [user]`
 - Lists the last established connections... either finished or not

File monitoring

File activity monitoring

- `fuser <filename>`
 - Identifies the processes being used by a file
- `lsof [filename | directory name]`
 - Lists open files

Disk activity

Used space

- `du [filename | directory name]`
 - Indicates used space per directory (including subdirs)

Free space

- `df [filename | directory name]`
 - Free space on each partition

I/O activity

- `iostat`
 - Per process and overall I/O statistics

Example top

```
top - 10:01:50 up 4 days, 8:40, 5 users, load average: 1.77, 1.51, 1.56
Tasks: 281 total, 1 running, 279 sleeping, 0 stopped, 1 zombie
%Cpu0  : 13.2 us,  3.3 sy,  0.0 ni, 82.9 id,  0.3 wa,  0.0 hi,  0.3 si,  0.0 st
%Cpu1  : 10.2 us,  1.5 sy,  0.0 ni, 87.3 id,  0.3 wa,  0.0 hi,  0.6 si,  0.0 st
%Cpu2  : 12.7 us,  1.5 sy,  0.0 ni, 84.6 id,  0.6 wa,  0.0 hi,  0.6 si,  0.0 st
%Cpu3  : 16.3 us,  1.7 sy,  0.0 ni, 81.6 id,  0.0 wa,  0.0 hi,  0.3 si,  0.0 st
KiB Mem : 16314076 total, 5436464 free, 3590272 used, 7287340 buff/cache
KiB Swap: 16360444 total, 16318936 free, 41508 used. 10859404 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
17901	rserral	1	0	1429512	265436	126648	S	16.5	1.6	4:51.75	slack
17115	rserral	5	0	2640856	349772	137352	S	9.6	2.1	5:00.66	gnome-shell
17340	rserral	1	0	1667320	157220	91880	S	4.6	1.0	0:33.14	slack
444	root	-51	0	0	0	0	S	2.0	0.0	17:17.13	irq/17-i2c_desi
17133	rserral	1	0	562520	236400	201880	S	1.7	1.4	0:51.53	Xwayland
17343	rserral	1	0	471912	48636	30472	S	1.7	0.3	0:00.92	python2
18210	rserral	1	0	3021200	577976	253764	S	1.3	3.5	4:42.75	firefox
286	root	-51	0	0	0	0	S	1.0	0.0	8:01.12	irq/17-idma64.1
20211	rserral	6	0	46988	3904	3044	R	1.0	0.0	0:00.33	top
19472	root	1	0	0	0	0	S	0.7	0.0	0:11.71	kworker/u8:2
6	root	1	0	0	0	0	S	0.3	0.0	13:19.49	ksoftirqd/0
7	root	1	0	0	0	0	S	0.3	0.0	2:02.42	rcu_preempt
17	root	1	0	0	0	0	S	0.3	0.0	13:23.78	ksoftirqd/1
23	root	1	0	0	0	0	S	0.3	0.0	14:30.76	ksoftirqd/2
29	root	1	0	0	0	0	S	0.3	0.0	16:11.32	ksoftirqd/3
621	message+	1	0	48732	6700	3072	S	0.3	0.0	4:09.41	dbus-daemon

Exercise

We have a database server with 1 CPU (and hyperthreading)

- Which is the problem present on the server if any?
- Which actions would you take?

```
top - 09:38:09 up 1 day, 18:29, 6 users, load average: 4.08, 4.93, 4.39
Tasks: 425 total, 12 running, 413 sleeping, 0 stopped, 0 zombie
%Cpu(s): 91.0 us, 6.8 sy, 0.9 ni, 1.3 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
KiB Mem : 16355660 total, 125088 free, 6559812 used, 9670760 buff/cache
KiB Swap: 33691644 total, 33689476 free, 2168 used. 8286212 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
4102	pcomp	20	0	2920500	1.029g	98884	S	46.1	6.6	103:32.24	firefox-esr
12802	pcomp	20	0	102332	68188	14164	R	30.6	0.4	0:00.93	chrome-bg-proc
12818	pcomp	20	0	80856	51980	17732	R	22.4	0.3	0:00.68	chrome-bg-proc
12835	pcomp	20	0	88840	49892	10524	R	17.1	0.3	0:00.52	chrome-bg-proc
3947	pcomp	20	0	2207552	505540	69276	S	14.5	3.1	49:25.10	gnome-shell
12861	pcomp	20	0	75972	37808	10480	R	12.2	0.2	0:00.37	chrome-bg-proc
12834	pcomp	20	0	65460	25816	8488	R	11.2	0.2	0:00.34	chrome-bg-proc
12873	pcomp	20	0	69680	32032	10508	R	9.2	0.2	0:00.28	chrome-bg-proc
12858	pcomp	20	0	59056	18824	8452	R	7.6	0.1	0:00.23	chrome-bg-proc
12833	pcomp	20	0	14312	11436	1356	R	6.9	0.1	0:00.21	mysqld

Exercise

We have a server with 32 logical CPU

- Which is the problem present on the server?
- How would you solve it?

```
top - 16:31:15 up 3:04, 20 users, load average: 29.76, 17.88, 10.19
Tasks: 1016 total, 2 running, 1013 sleeping, 1 stopped, 0 zombie
Cpu(s): 2.5%us, 1.2%sy, 0.0%ni, 86.8%id, 9.4%wa, 0.0%hi, 0.1%si, 0.0%st
Mem: 65969572k total, 33193236k used, 32776336k free, 8656k buffers
Swap: 16777208k total, 7635416k used, 9141792k free, 31292k cached
PID  USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
3164  tst8       20   0 23.1g 21g  584  R 100.0 34.1   7:44.76 emacs
4576  tst8       20   0 104m 1080  476  S  53.3  0.0   2:17.90 genarray.sh
1010  root       20   0      0      0      0  D   2.0  0.0   2:07.06 kmirrord
3342  g_users    20   0 15868 1528  476  R   1.0  0.0   1:43.80 top
168   root       20   0      0      0      0  S   0.3  0.0   0:02.09 events/21
2568  tst6       20   0 101m  376  240  S   0.3  0.0   1:27.30 sshd
```

Outline

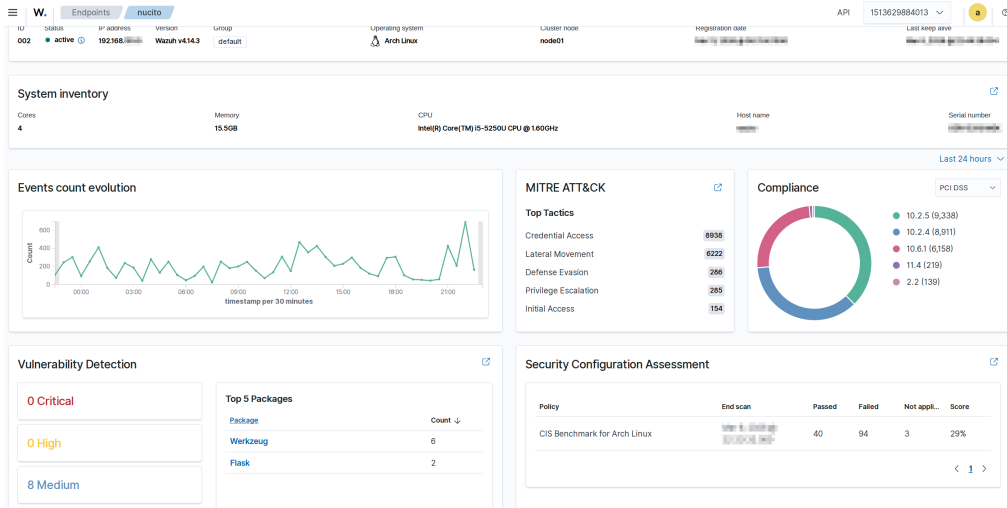
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Network monitoring

Integrated systems

- Centralized information for various servers
 - Resources
 - Services
 - Uptime
 - Connectivity
 - Logs
- Ease the issue detection
- Wazuh, Splunk, ...

Example: Wazuh



Personal homework

Filesystems

- ext4
- btrfs
- xfs
- zfs
- ntfs

Backup tools

- Bacula
- borg
- tar
 - gzip
 - bzip2